Venkata Lakshmi Parimala Pasupuleti

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github.com/venkatalakshmiparimala/AgentProbe | San Francisco, CA / New York, NY

PROFESSIONAL SUMMARY

AI Engineer with 9 years of production Python engineering, enterprise-scale delivery (1.6M+ financial records, 100K+ MAU), and hands-on agentic AI deployment. Built **AgentProbe** - a 3-agent LangGraph runtime with production-grade RAG, multi-provider model routing, and CI/CD quality gates - deployed as a 15-endpoint FastAPI backend. Applied AI at enterprise scale across regulated financial and healthcare data with output validation, prompt engineering, and eval frameworks. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

WORK EXPERIENCE

George Mason University
Aug 2024 -- Aug 2025

Research Associate -- Virginia, USA

- Built regression models ($R^2 = 0.977$) in Python and R across WHO and UK Road Safety datasets spanning 190+ countries; applied Isolation Forest to automate data-quality checks across 3 research pipelines.
- Designed Tableau, ggplot2, and Quarto dashboards for faculty leadership -- improved research comprehension by 40% across two cohorts.

Entrata
Jan 2017 -- Aug 2024

Software Engineer

- Applied Python, PyTorch, and OpenAI API to build **AI data transformation pipelines across 1.6M financial records** for enterprise property management clients -- production AI on regulated data with output validation, quality gates, and compliance controls.
- Reduced deployment times 40% building 20+ microservices (Python, PostgreSQL, RabbitMQ) across 120+ Agile sprints for **100K+ monthly active users** at enterprise scale; cut execution time 50% via Docker containerization; improved query performance 30% across 200+ SQL queries.
- Achieved 100% PHI compliance migrating all client data to AWS S3 with Libsodium-encrypted pipelines -- zero security incidents; built affordable housing compliance modules (Section 8/HCV, LIHTC, HUD) - income eligibility verification, AMI threshold calculations, and HUD regulatory reporting across 100K+ unit portfolios; eliminated 80% of manual operational workload via Python/PL-SQL automation.

PROJECTS

AgentProbe: Agentic AI Testing & Eval Toolkit
Open Source

Python · LangChain · LangGraph · FastAPI · ChromaDB · OpenAI · Anthropic · Groq · Ollama · React · Docker · GitHub Actions -- github.com/venkatalakshmiparimala/AgentProbe

- Built a **3-agent LangGraph runtime** (Orchestrator, RAG Tracer, Auto Fix) with tool calling, inter-agent state handoffs, and multi-provider model routing (Anthropic, OpenAI, Groq, Ollama) -- deployed as a 15-endpoint FastAPI backend running 100% locally.
- Automated production-grade quality gates with a **5-metric LLM-as-judge eval engine** (factual accuracy, hallucination, semantic retrieval, completeness, compliance) and CI/CD integration; React dashboard tracking RAG retrieval quality, latency, and regression drift across concurrent agent test suites.

AI on FHIR: NLP-Driven Medical Records Query System

Flask · LLMs · FHIR · React · spaCy · BERT · Embeddings · Docker -- github.com/venkatalakshmiparimala/AI_on_FHIR

- Reduced clinical retrieval time ~60% by building an NLP platform translating natural language queries (EN/ES) to FHIR APIs using LLMs, BERT, and semantic retrieval -- **HIPAA-compliant RBAC** across structured healthcare records.

PropLens: Lease Intelligence & Compliance AI
In Progress

Python · LangGraph · ChromaDB · FastAPI · LLMs · React

- Building agent-based RAG pipeline over lease PDFs -- LangGraph agents + ChromaDB vector search for compliance automation across 1,000+ tenant records; LLM-as-judge scoring with anomaly detection.

EDUCATION

M.S. Data Analytics Engineering -- George Mason University — Graduated May 2026

GPA: 3.87 · Fairfax, Virginia

B.Tech Computer Science & Engineering -- Jawaharlal Nehru Technological University, Anantapur — 2013 -- 2017

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, R, Bash, HTML/CSS

AI & Agents: LangChain, LangGraph, Multi-Agent Orchestration, RAG Pipelines, Vector Search, Embeddings, Prompt Engineering, LLM Evaluation, LLM-as-Judge, Eval Frameworks, Agent Observability, OpenAI API, Anthropic API, Groq, Ollama, Hugging Face, FastAPI

ML & Data: PyTorch, TensorFlow, Scikit-learn, XGBoost, Isolation Forest, BERT, spaCy, NumPy, Pandas, Tableau, ggplot2

Cloud & Infrastructure: AWS (S3, EC2, RDS, Glue), Azure, GCP, Docker (production containerization), Kubernetes (orchestration patterns), CI/CD (GitHub Actions), RabbitMQ, PostgreSQL, MySQL